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Data Analysis Essentials

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Module

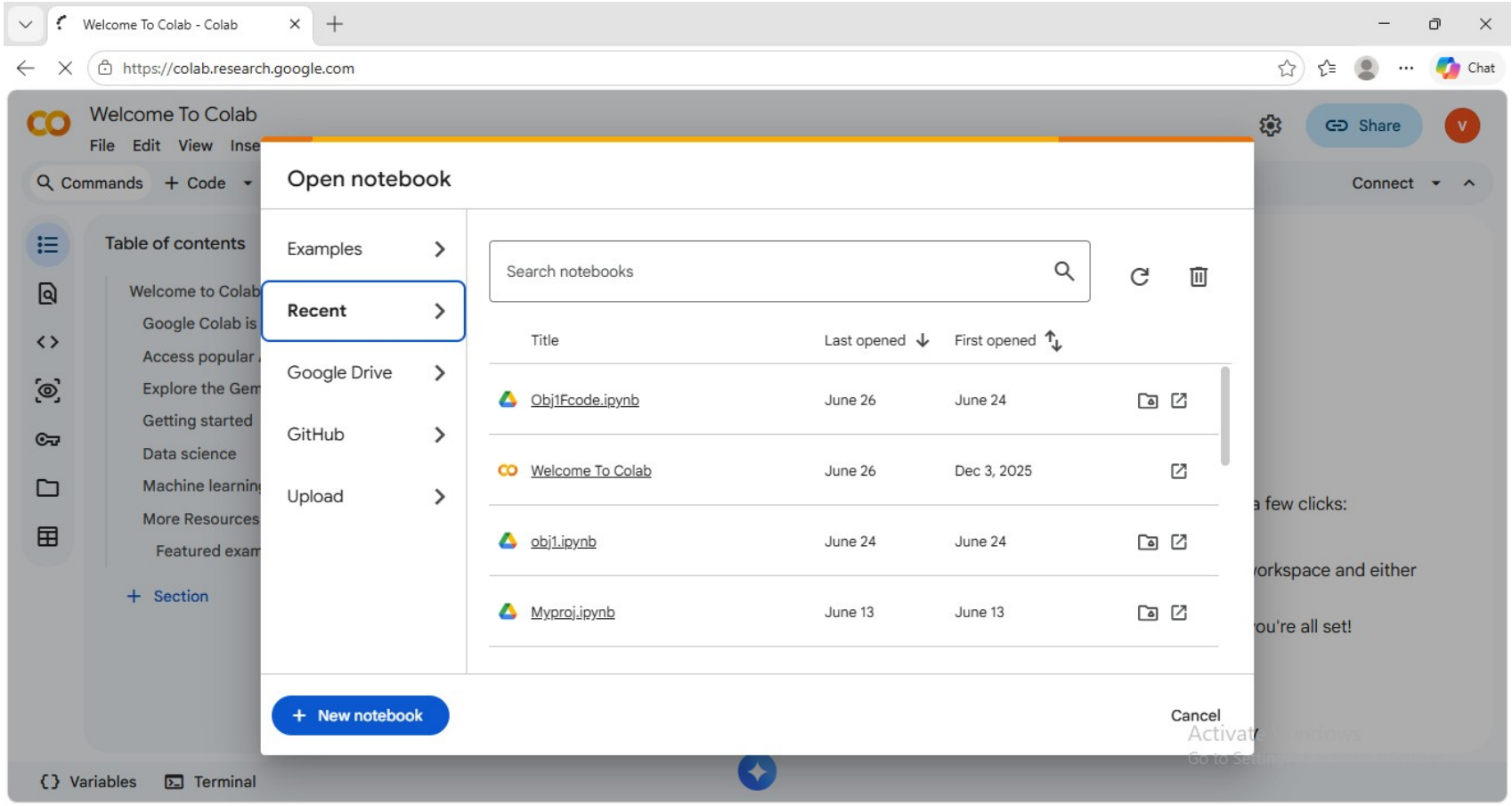
- Module is a file with a .py extension that has a set of reusable classes,functions and variables. You want to use functions or variables defined in the module will simply import that particular module.
- To use a module is to add import module_name as the first line of your program.

We will discuss about different modules

- numpy
- pandas
- matplotlib

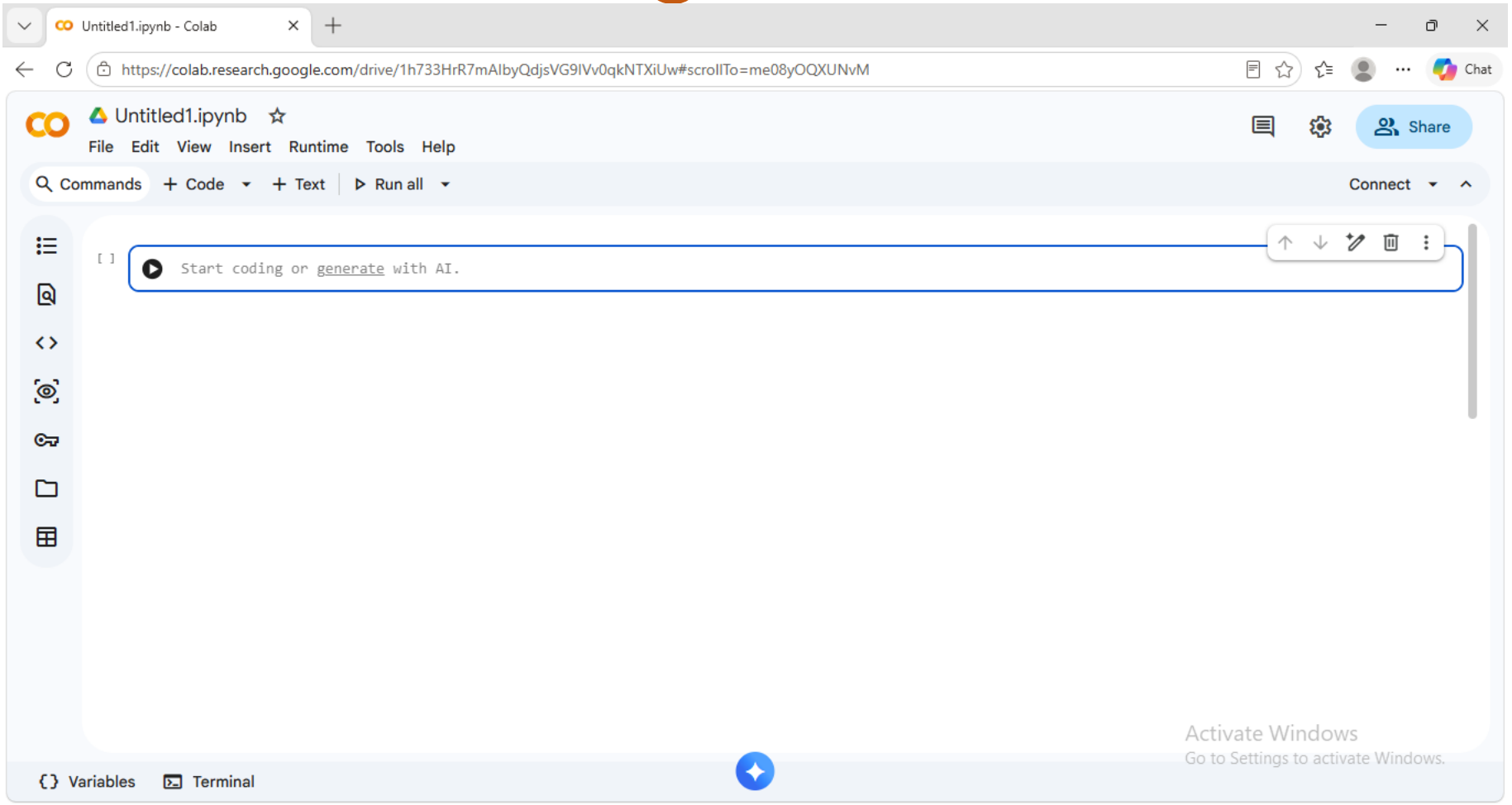
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NUMPY

What is NumPy?

NumPy (Numerical Python) is a Python library used for numerical computing. It provides a powerful N-dimensional array object (ndarray) and functions to perform mathematical, statistical, and matrix operations efficiently.

Advantages of NumPy

- Faster than Python lists
- Less memory usage
- Supports multidimensional arrays
- Provides mathematical and linear algebra functions
- Widely used in Data Science, Machine Learning, and Artificial Intelligence

Installation

pip install numpy

Import NumPy Module

import numpy as np

Lab Practice

Practice:

Week 1: Creating a NumPy Array a. Basic ndarray

- b. Array of zeros
- c. Array of ones
- d. Random numbers in ndarray
- e. An array of your choice
- f. Imatrix in NumPy
- g. Evenly spaced ndarray

ndarray

An ndarray (N-dimensional array) is the primary data structure in NumPy. It stores elements of the same data type in a fixed-size array.

Syntax

np.array(object)

a) Program

```
import numpy as np

arr = np.array([10, 20, 30, 40, 50])

print("Array:")

print(arr)print("Type:", type(arr))

print("Shape:", arr.shape)

print("Data Type:", arr.dtype)
```

Output

```
Array: [10 20 30 40 50]

Type: <class 'numpy.ndarray'>

Shape: (5,)

Data Type: int64
```

ndarray

(b) Array of Zeros

The zeros() function creates an array in which every element is *0*.

Syntax

np.zeros(shape)

Program

```
import numpy as np
arr = np.zeros((3,4))
print(arr)
```

Output

```
[[0. 0. 0. 0.]
[0. 0. 0. 0.]
[0. 0. 0. 0.]]
```

ndarray

Program

```
import numpy as np
arr = np.zeros((2,3), dtype=int)
print(arr)
```

Output

```
[[0 0 0]
[0 0 0]]
```

ndarray

(c) Array of Ones

The ones() function creates an array where every element is 1.

Syntax

np.ones(shape)

Program

```
import numpy as np
arr = np.ones((3,3))
print(arr)
```

Output

```
[[1. 1. 1.]
[1. 1. 1.]
[1. 1. 1.]]
```

ndarray

Integer Ones

Program

```
import numpy as np
arr = np.ones((2,5), dtype=int)
print(arr)
```

Output

```
[[1 1 1 1 1]
[1 1 1 1 1]]
```

ndarray

(d) Random Numbers in ndarray

NumPy provides several functions to generate random numbers.

Program

```
import numpy as np
arr = np.random.rand(3,3)
print(arr)
```

Output

```
[[0.46920317 0.38972024 0.37604085]
[0.66665463 0.77663535 0.21276047]
[0.47222453 0.14646436 0.36144166]]
```

ndarray

ii) Random integers

Program

```
import numpy as np
arr = np.random.randint(1,100,size=(3,4))
print(arr)
```

Output

```
[[ 3 89 72 37]
[51 52 56 12]
[77 7 81 24]]
```

ndarray

iii) Random values from normal distribution

Program

```
import numpy as np
arr = np.random.randn(3,3)
print(arr)
```

Output

```
[[-2.14430461 -2.2649812 -0.07597307]
[ 0.48873054 0.16615132 0.99582568]
[-0.72180803 -0.6550189 1.0016304 ]]
```

ndarray

(e) An Array of Your Choice

You can create arrays manually by entering the desired values.

Program

```
import numpy as np
student_marks = np.array([85, 92, 78, 88, 95])
print("Student Marks:")
print(student_marks)
```

Output

```
Student Marks: [85 92 78 88 95]
```

2Darray

2D Array Example

Program

```
import numpy as np
matrix = np.array([    [10,20,30],    [40,50,60],    [70,80,90]])
print(matrix)
```

Output

```
[[10 20 30]
[40 50 60]
[70 80 90]]
```

ndarray

(f) Identity Matrix in NumPy

An Identity Matrix is a square matrix in which: All diagonal elements are 1

All other elements are 0

NumPy provides the eye() function.

Syntax

np.eye(n)

ndarray

Program

```
import numpy as np
identity = np.eye(4)
print(identity)
```

Output

```
[[1. 0. 0. 0.]
[0. 1. 0. 0.]
[0. 0. 1. 0.]
[0. 0. 0. 1.]]
```

ndarray

(g) Evenly Spaced ndarray

The arange() and linspace() functions are used to generate evenly spaced numbers.

Syntax

np.arange(start, stop, step)

Program

```
import numpy as np
arr = np.arange(0,20,2)
print(arr)
```

Output

[0 2 4 6 8 10 12 14 16 18]

Using linspace()

```
import numpy as np
arr = np.linspace(0,10,6)
print(arr)
```

Output

[0. 2. 4. 6. 8. 10.]

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Thank You